Response to Reviewers

Manuscript NEURI-D-26-00299, *Neuroscience Informatics*. The manuscript has been re-analysed and rewritten (new title: "An Audited Evaluation of Explainable RF–XGBoost Stacking for Multiclass Sleep-Disorder Classification on a Small Public Benchmark"). Section, table and figure numbers refer to the revised manuscript.

Summary of principal changes

- **Outcome coding verified.** The 219 "no disorder" records carry the literal label "None"; the earlier "missing" description was a pandas import artefact (Section 3.1).
- **Duplicates audited.** 242 exact duplicates (132 unique rows, 109 predictor vectors); 22 vectors with conflicting labels cap accuracy at 93.3% (Section 3.2).
- **One model throughout.** All analyses evaluate the primary RF–XGB stack; the RF–XGB–SVM variant has been removed. The SVM is only an independent baseline.
- **New validation.** Repeated 10×5-fold grouped CV, repeated nested CV with tuning confined to training partitions, unique-row sensitivity analysis, 13 learners including logistic regression and a shallow tree, paired tests, calibration (Sections 4–5).
- **Headline claim withdrawn.** The reported 97.33% (73/75) could not be reproduced (we obtain 72/75), and 68 of 75 test records duplicate training records. Grouped CV gives 89.6% for the stack, with no advantage over soft voting, its base learners or logistic regression.
- **Explanation audit.** SHAP stability over 10 independent runs, collinearity analysis, audited LIME with identifiers and fidelity.
- **Released:** code, software versions, seeds, split identifiers, observation-level predictions.
- **Condensed:** Table 1, the ANN comparison table, PCA, the ALE surface, ROC curves and the repeated clinical caveats were removed or merged.

Reviewer #1

1. Verify outcome coding.

Response. Verified against the source CSV: the column contains exactly the strings `None` (219), `Sleep Apnea` (78) and `Insomnia` (77), with no empty fields. The default pandas parser turns `None` into missing values, which produced the earlier description. The analysis is now run with `keep_default_na=False` and asserts the labels (Section 3.1; code `common.py`). No repetition of the analysis under a different coding was needed because the labels were not missing; the meaning of "None" is discussed as a limitation (Section 7).

2. Reconcile model configurations.

Response. Done. Every analysis now uses the two-learner RF–XGB stack of Algorithm 1; the RF–XGB–SVM stack, the ablation figure and the seed-sweep figure are removed. All hyper-parameters are listed in Appendix A.

3. Reproducibility.

Response. The supplementary archive contains executable code, software versions (Section 4.5), per-repeat seeds, split identifiers for every protocol, and a script that regenerates all tables and in-text numbers. Pre-processing (scaler, encoder vocabulary) is refitted inside every inner fold (Algorithm 1); outer test folds

are never used for fitting or selection. Our re-implementation of the original single split gives 72/75 (not 73/75); we say so explicitly in Section 5.1 and withdraw the earlier figure.

4. Strengthen validation and comparisons.

Response. Sections 4.2 and 5.2–5.4: repeated stratified-grouped 10×5-fold CV and repeated nested CV (inner 3-fold selection on training partitions) compare stacking, soft voting, RF, XGBoost, logistic regression, a depth-3 tree, majority class and six further learners on identical splits, with variability, paired differences and corrected tests (Tables 2 and 4, Figs. 2 and 3). The stack is not distinguishable from soft voting (-0.2 pp, 95% CI -1.8 to +1.1).

5. Verify statistical results.

Response. Observation-level labels, predictions and probabilities for every model, repeat and fold are provided, together with the single-split output; McNemar tests and bootstrap intervals are computed from them. On the re-implemented single split the stack and soft voting both score 72/75 (paired CI -4.0 to 4.0 pp). Conclusions now consistently state that there is no evidence of superiority, and the discussion explains the limited power (Section 6).

6. Explanation reporting.

Response. Table 8 reports Person ID, reference label, predicted probabilities, LIME surrogate R^2 and seed-to-seed rank stability. Explanation sampling is specified (25-record background from training; 35 explanation records from the test partition; $2p+1$ evaluations) and repeated over 10 independent runs (Table 7, Fig. 5): within-model rank correlation 0.98 (stack) and 0.97 (soft voting). The shaded bands of the old repeated-split figure no longer exist because the figure was removed; all intervals are now defined in Section 4.3.

Methodological contribution.

Response. We no longer describe a "proposed framework". Section 6 states that stacking, SHAP and LIME are standard and that the contribution is a verified implementation, a quantified account of duplicate and single-split inflation, like-for-like comparisons with paired statistics and calibration, and an explanation audit.

Reviewer #2

1. Limited novelty.

Response. Agreed; see the response above. The title, abstract, introduction and discussion are reframed as an audit and evaluation rather than a new method.

2. Clinical interpretation limited by data and outcome definition.

Response. The task is now described as classification of a recorded status, not risk stratification or decision support (title, keywords, Section 3, Section 6 "Benchmark outcomes versus clinical screening", Section 7).

3. High performance on a small hold-out requires rigorous validation.

Response. Addressed by repeated grouped and nested CV with cluster-bootstrap intervals. The single-split figure is shown to be inflated by duplicates (Section 5.1); performance on unique rows is 73.0% (Section 5.2).

4. Explanation methods do not demonstrate mechanisms.

Response. Stated explicitly in Sections 4.4 and 6; PCA and ALE were removed, and the remaining analyses are framed as audits of fitted functions with stability and collinearity checks.

5. Comparison with recent competitive tabular approaches.

Response. We added extra trees, HistGradientBoosting, LightGBM, CatBoost, an RBF SVM and an MLP, plus logistic regression and a shallow tree (Table 2, Fig. 2). Tabular foundation models and attention-based

tabular networks could not be run because their pretrained weights were not accessible in our environment; this is stated as a limitation (Section 7).

6. Presentation, formatting, figures, references.

Response. The manuscript was rewritten in a consistent structure. All equations were re-typeset; all figures and tables are regenerated programmatically from stored results (no AI-generated artwork). The reference list was rebuilt and unused entries removed.

Reviewer #3

1. Separate benchmark outcomes from clinical screening.

Response. Added a dedicated paragraph in Section 6.

2. Expand interpretation of McNemar/bootstrap.

Response. Section 6 ("Statistical power"): two discordant records; minimum detectable difference about 5.3 points at 80% power; about 1180 test records would be needed to detect a 1.33-point difference.

3. Collinearity and attribution differences between architectures.

Response. Section 5.7: Spearman and variance-inflation factors; stack versus soft-voting attributions correlate weakly ($\rho = 0.18$), but total blood-pressure attribution is almost identical (0.148 versus 0.156) and joint permutation of the pair matters far more than either member alone.

4. Repeated-split analysis used a different model.

Response. Removed; the repeated analyses now evaluate the primary RF–XGB stack (Section 5.2), and the manuscript states this in the protocol description.

5. Labelling note in limitations.

Response. Added (Section 7, item ii), including the possible bias towards under-detection.

6. Local explanation reporting.

Response. Table 8 reports the identifiers, labels, probabilities and fidelity, and Section 8 lists the information future local-explanation figures should report.

7. Future work.

Response. Section 8 recommends repeated CV, calibration assessment and external clinical validation. The first two are also performed here (Sections 5.2–5.5).

8. Suggested references.

Response. We read the three suggested works. They concern traffic-flow forecasting, image semantic segmentation and remote-sensing object detection; we could not identify a claim in the revised manuscript that they would support, so we have not cited them, and we are glad to add them if the editor considers them relevant. We expanded the reference list with works directly relevant to evaluation, calibration, tabular benchmarks and explanation.

Reviewer #4

Major 1. Provenance, synthetic data, "None" coding.

Response. Coding verified (see R1.1). Provenance: no collection protocol is documented and the duplicate structure suggests templated generation; the manuscript now says so and treats the data as a methodological benchmark (Section 3.1).

Major 2. "Supplied" results, no code.

Response. All results are now generated by the released code; the word "supplied" no longer appears, and code, split indices and predictions are provided.

Major 3. Duplicates and tuning.

Response. Duplicates are reported (242) and handled by grouped splitting; unique-row results are given. The original settings were fixed values of unverifiable provenance, so we repeat the evaluation with nested tuning (Section 5.4). Results are unchanged in substance.

Major 4. Repeated nested CV, CIs, calibration, baselines.

Response. Done (Sections 5.2–5.5); logistic regression and a depth-3 tree are included. Logistic regression reaches 89.7% against 89.6% for the stack.

Major 5. Align explanation analyses; SHAP stability.

Response. Seed/ablation analyses and Fig. 9a were removed. SHAP and LIME are reported for both the stack and the soft-voting model with explicit model labels, and SHAP stability was assessed over 10 independent backgrounds and samples; collinearity is analysed (Section 5.7).

Minor 1. "Risk stratification", circularity.

Response. Terminology changed throughout. Section 5.6 shows that removing sleep duration and quality (and stress) leaves accuracy essentially unchanged.

Minor 2. Novelty/"proposed framework".

Response. Softened; see Section 6.

Minor 3. Table 6 (ANN accuracies on a different split).

Response. Removed.

Minor 4. Fig. 8 index/probability; Fig. 11 bands; train/test accuracy of 0.926/0.924.

Response. Local explanations are now audited (Table 8); Fig. 11 was removed. The closeness of training and test accuracy is explained by the accuracy ceiling (in-sample accuracy on all records is 93.0% for RF against the 93.3% ceiling) and by duplicated rows spanning the split (Section 5.1).

Minor 5. ALE grid.

Response. The ALE panel was removed.

Minor 6. Condense Related Work/Table 1/repeated caveats.

Response. Related Work is now one short section; Table 1 was removed; the clinical caveat is stated once in Discussion and once in Limitations.

Reviewer #5

1. 219 blank outcomes.

Response. See R1.1. The exact import, cleaning and recoding code is provided (`common.py`, function `load`).

2. 97.33% on 75 observations.

Response. The figure could not be reproduced and is withdrawn (Section 5.1); repeated CV replaces it (Sections 5.2–5.4). The perfect per-class results of the single split are no longer reported.

3. Code, versions, seeds.

Response. Provided (Section 4.5 and archive).

4. Inconsistent model in ablation/repeated analysis.

Response. Resolved (see R1.2).